

Detecting Faulty PCBs Using Thermal Image Analysis

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Introduction

Printed Circuit Boards (PCBs) form the backbone of modern electronic systems, supporting applications ranging from consumer electronics to safety-critical industrial platforms. As system complexity and power density continue to increase, early detection of PCB faults has become increasingly important to ensure operational reliability, prevent system failures, and reduce manufacturing and maintenance costs. Faulty PCBs may exhibit defects such as short circuits, open traces, poor solder joints, or degraded components--many of which manifest as abnormal thermal behavior during operation.

Traditional PCB inspection techniques often require direct line-of-sight or specialized equipment. These approaches can be time-consuming, costly, and limited in their ability to detect latent or thermally induced faults. Thermal imaging, by contrast, offers a non-contact and non-destructive method for observing heat distribution across a PCB under operating conditions, enabling the identification of subtle anomalies through temperature gradients and localized hotspots.

This project aims to detect faulty PCBs through thermal image analysis by extracting meaningful features from thermal images. The ultimate goal is to evaluate the effectiveness of thermal imaging in accurately predicting PCB faults and to assess the feasibility of deploying such methods in real-world inspection and quality assurance pipelines.

Data Source

Due to the lack of direct access to a thermal imaging camera and a large number of PCB samples, a synthetic dataset was generated for the purpose of this study. The dataset was designed to emulate the thermal characteristics of real PCBs under operational conditions.

Synthetic PCB thermal images were generated by randomly placing heat sources on a two-dimensional grid representing a PCB surface. Normal boards were modeled using smoothly varying temperature distributions, while faulty boards included localized regions of elevated temperature to simulate defects such as short circuits, degraded components, or abnormal current paths.

These fault regions were introduced at random spatial locations and varied in size and intensity to reflect a range of possible thermal behaviors.

Thermal images were encoded using color-mapped temperature values, where variations in color corresponded to differences in simulated temperature. Each generated image was labeled according to fault status and stored for downstream analysis. This approach enabled the creation of a sufficiently large and balanced dataset for training and evaluating machine learning models while maintaining control over fault placement and severity.

Although the dataset is synthetic, it provides a useful testbed for evaluating feature extraction methods

and classification performance. The methodology emphasizes algorithmic validation and pattern recognition rather than absolute temperature accuracy, serving as a preliminary step toward future deployment on real-world thermal PCB data.

Data Format

Sdf

Method Documentation and Reasoning

Synthetic Data Generation

Synthetic thermal PCB images were generated using `generateSyntheticPCB(imgSize, padGrid, padRadius, defectCoords)`. The PCB surface was modeled as a grid of circular pads to approximate repeated copper pad structures. For each pad location (i, j) , the code assigns a pixel intensity based on whether the pad is defective. Normal pads were assigned a baseline intensity, while defective pads were assigned higher intensity to represent increased heat dissipation typical of electrical faults such as shorts or abnormal resistance. Random defect coordinates were sampled using `randi()` to vary defect placement across samples.

To emulate realistic thermal sensor behavior, additive Gaussian noise was applied to each generated image. Although this dataset is synthetic, controlled defect injection provides a consistent testbed for validating hotspot detection logic and quantifying whether the algorithm identifies the intended anomalous regions.

Preprocessing

Each PCB image was loaded and converted to grayscale using `mat2gray()`, ensuring pixel values are normalized into a comparable intensity range. A Gaussian blur (`imgaussfilt`) was applied to reduce high-frequency

noise and suppress isolated pixel artifacts prior to thresholding. This step improves stability of downstream binary segmentation by preventing spurious hotspot detection.

Hotspot Segmentation and Region Extraction

Hotspots were detected through intensity thresholding using a fixed threshold parameter (`threshold = 0.8`). Pixels above the threshold were classified as candidate hotspot pixels, producing a binary mask. Small connected regions were removed using `bwareaopen(..., 10)` to eliminate noise-driven detections below a minimum size. Connected hotspot regions were then analyzed using `regionprops()` to extract two key geometric descriptors:

- Centroid: estimated hotspot location
- Area: hotspot size in pixels

These outputs provide interpretable thermal anomaly indicators that can later be used for feature-based fault classification, quality scoring, or defect localization.

Visualization and Reporting

For each PCB instance, the code generates an annotated visualization by overlaying the detected hotspot centroids on the normalized thermal image using a heat colormap. A report is automatically compiled into a single PDF using MATLAB Report Generator (`mlreportgen.dom.*`). Each PCB section contains a side-by-side comparison of

- the raw thermal image
- the hotspot-annotated image

In addition, CSV and text summaries are written for structured logging and later statistical analysis. Finally, intermediate files are deleted to ensure only the final compiled PDF remains as the primary artifact.

Findings Recording and Analysis

Findings were recorded per PCB sample in both visual and numerical formats to support traceability and evaluation. The hotspot detection pipeline outputs (i) the number of detected hotspot regions, (ii) their centroid coordinates, and (iii) their region areas. These metrics provide a consistent method for quantifying anomaly count and severity across samples.

Qualitative validation is supported through the automatically generated annotated thermal overlays, which enable inspection of whether detected centroids align with the injected defective pad locations. Quantitative validation is supported through the exported CSV tables, allowing aggregation of hotspot statistics across the dataset (e.g., mean hotspot count per board, area distributions, and spatial clustering trends). Because the defect positions are injected programmatically, the dataset provides an internal ground-truth reference for verifying detection correctness in a controlled environment.

Limitations and Future Work

This study relies on synthetically generated thermal PCB images rather than data collected from physical hardware. While the synthetic dataset enables controlled experimentation and clear validation of hotspot detection logic, it does not capture all sources of variability present in real-world thermal imaging, such as emissivity differences between materials, environmental heat transfer effects, camera calibration errors, or non-uniform sensor noise. As a result, the absolute temperature values and visual

patterns may differ from those observed in real PCB thermal scans.

Additionally, defect detection in this work is performed using fixed threshold-based segmentation. Although this approach is effective for identifying clearly defined thermal anomalies in a controlled environment, it may be sensitive to parameter selection and less robust when applied to boards with complex thermal profiles or uneven baseline heating. More advanced segmentation or learning-based approaches may improve generalization and adaptability.

Future work will focus on validating the proposed methodology using real thermal images acquired from physical PCBs under controlled electrical loading. Incorporating adaptive or machine learning-based detection models, such as convolutional neural networks or unsupervised anomaly detection techniques, may further improve fault classification accuracy. Expanding the dataset to include a broader range of defect types and severities would also strengthen model robustness and enable more comprehensive performance evaluation.